

What is Retrieval-Augmented Generation?

Retrieval-augmented generation, or RAG, combines a retrieval step with a generation step. When a user asks a question, the system first searches an external knowledge source, such as a folder of documents or a vector database, for passages relevant to the question. Those passages are then inserted into the prompt given to a language model, which generates an answer grounded in the retrieved text rather than relying solely on what it learned during training.

This approach has two main advantages over asking a language model a question directly. First, it lets the system answer questions about information the model was never trained on, such as private company documents or content published after the model's training cutoff. Second, it reduces hallucination, because the model is instructed to base its answer on the retrieved passages, and those passages can be cited so the user can verify the answer against the source.

RAG systems are not without failure modes. If the retrieval step fails to surface the right passages, the generation step has nothing good to work with, no matter how capable the underlying model is. This is why evaluating the retrieval step separately from the generation step, using frameworks such as RAGAS, is an important part of building a reliable RAG system.